HDF5: API Specification
Reference Manual

The HDF5 library provides several interfaces, each of which provides the tools required to meet specific aspects of the HDF5 data-handling requirements.

See below for the FORTRAN90 and C++ APIs.

Library Functions	The general-purpose **H5** functions.
Attribute Interface	The **H5A** API for attributes.
Dataset Interface	The **H5D** API for manipulating scientific datasets.
Error Interface	The **H5E** API for error handling.
File Interface	The **H5F** API for accessing HDF files.
Group Interface	The **H5G** API for creating physical groups of objects on disk.
Identifier Interface	The **H5I** API for working with object identifiers.
Property List Interface	The **H5P** API for manipulating object property lists.
Reference Interface	The **H5R** API for references.
Dataspace Interface	The **H5S** API for defining dataset dataspace.
Datatype Interface	The **H5T** API for defining dataset element information.
Compression Interface	The **H5Z** API for compression.
Tools	Interactive tools for the examination of existing HDF5 files.
Predefined Datatypes	Predefined datatypes in HDF5.
Ragged Arrays	The **H5RA** API for ragged arrays was removed from the HDF5 library at Release 1.4.

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The Fortran90 and C++ APIs to HDF5

The HDF5 Library distribution now includes FORTRAN90 and C++ APIs. These APIs are described in the following documents.

Fortran90 API

HDF5 FORTRAN90 User's Notes contains general information regarding the API. Specific information on each API call is found in the *Reference Manual*.

Fortran90 APIs in the *Reference Manual*: The current version of the *HDF5 Reference Manual* includes a description of the FORTRAN90 API to HDF5. Fortran subroutines exist in the H5A, H5D, H5E, H5F, H5G, H5I, H5P, H5R, H5S, and H5T interfaces and are described on those pages.

In general, each Fortran subroutine performs exactly the same task as the corresponding C function. The links at the top of each reference manual section go to the C function descriptions, which serve as general descriptions for both the C and Fortran APIs. A button, under **Non-C API(s)** at the end of the C function description, opens an external browser window displaying the Fortran-specific information. You will probably want to adjust the size and location of this external window so that both browser windows are visible and to facilitate moving easily between them.

HDF5 Fortran90 Flags and Datatypes lists the flags employed in the Fortran90 interface and contains a pointer to the HDF5 Fortran90 datatypes.

C++ API

HDF5 C++ User's Notes

HDF5 C++ Interfaces

(Note that the C++ APIs are not yet integrated into the *HDF5 Reference Manual*.)

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